

Day 176/365

Random Forest: where it shines and where it struggles

Applications, advantages and limitations in
simple words.

Quick refresher

A random forest is **many decision trees**, each trained on different random rows, whose answers are **averaged** (for numbers) or **voted** (for categories).

Over the last 3 days I built one and found its catch on a tiny dataset. Today: the bigger picture.

1. Where is it used?
2. What is it good at?
3. Where does it struggle?

Where it's used

Applications

Predicting numbers

Anything where the answer is a quantity.

Example: the price of a house, or how much a customer will spend over the years.

Risk analysis

Finding which factors push risk up, in areas like healthcare and finance.

Example: which details about a loan applicant matter most for repayment.

Where it's used

Applications, continued

Lots of input columns

It copes well when each row has dozens or hundreds of features.

Example: 200 columns describing how a customer uses an app.

Curved, tangled relationships

When the answer doesn't change in a straight line.

Example: a flat's price doesn't rise evenly with size. Location and age change the pattern.

What it's good at

Advantages (1 of 2)

Handles curves

It can learn bends and jumps without you adding x^2 columns, as I had to do with polynomial regression.

Overfits less than one tree

One deep tree memorizes. 100 different trees averaged are steadier, like Student A from my Day 168 post.

Not thrown off by outliers

Say one mansion has an absurd price. A single tree may build a whole branch around it. In a forest, it influences only some trees, and the average dilutes it.

What it's good at

Advantages (2 of 2)

Works with big data

Many rows and many columns are fine.

Copes with some missing values

The article lists this as a strength. It depends on your library version, so check before relying on it. Older versions need you to fill gaps first.

No scaling needed

Trees ask "is it above this number?", so units don't matter. In my car price series I had to standardize for Ridge. Not here.

Where it struggles

Limitations (1 of 2)

Slow to train

100 trees take 100 times the work of one.

What helps: fewer trees, or train them in parallel.

Hard to explain

You can't read a forest like a single flowchart or like linear regression coefficients.

What helps: check which features it relies on most (feature importance).

Uses a lot of memory

All those trees have to be stored.

What helps: limit tree depth.

Limitations (2 of 2)

Noisy data can still fool it

If the data is full of random mistakes, trees can still memorize them.

What helps: limit depth and leaf size, as in my decision tree Part 3.

Imbalanced data

If 95 of 100 patients are healthy, it may just favour "healthy". This hits classification most.

What helps: balance the classes, and judge with precision and recall.

Can't predict beyond what it has seen

My Day 175 lesson. Its answers are averages of values from training, so it never goes above the highest one.

Not in the article, but important for forecasting.

Takeaways

A strong all-rounder with a few clear weak spots.

Reach for it when the data is large or messy, relationships are curved, and you don't need to explain every detail.

Think twice when you must explain each prediction, forecast beyond the data, or have very little data.

Which one would you try first on a house price dataset: linear regression or a random forest?